

Terra Firma Partners — Project Instructions

Overview

Terra Firma is a hunting/land intelligence platform focused on Missouri. Business model centers on \$19 parcel unlocks and Pro (\$99/yr) / Pro Max (\$199/yr) subscriptions.

Active Product SKUs (April 2026)

- **\$19 Parcel Unlock** — one-time purchase via `/api/parcels/purchase`, unlocks parcel for download. `/intel` page shows “Hunt Report” as the downloadable PDF artifact.
- **\$99/yr Pro subscription** — recurring via `/pricing` + `/api/stripe/checkout`.
- **\$199/yr Pro Max subscription** — recurring via `/pricing` + `/api/stripe/checkout`.

Removed Products

- **\$149 Hunt Intelligence Report** — fully removed April 24, 2026 (Session 89). `hunt_report` `productType` no longer exists.
- **\$49 Land Intelligence Report / Broker Quick Look** — fully removed April 24, 2026 (Session 90). `land_report` `productType` no longer exists. `/api/orders` POST and `/api/checkout` return 410 Gone. `/map` page is now a parcel viewer only (no checkout). `/brokers` page promotes subscriptions instead. `/checkout/success` is a generic thank-you page.
- Do NOT reintroduce either product without user approval.

Architecture Notes

- Stripe is in LIVE mode — never create/modify Stripe products or prices.
- `lib/report-generators.ts` was DELETED along with the \$49 product (it only contained `generateLandPdfDirect`). `generateHuntPdfDirect` was deleted with \$149 removal.
- `/api/parcel-hunt-file/route.ts` is still used for the \$19 parcel unlock flow — KEEP IT.
- `/api/generate-pdf/route.ts` default branch (12-page PDF) preserved for any legacy order download attempts.
- `/intel` page references to “Hunt Report” refer to the downloadable PDF unlocked by the \$19 parcel purchase, NOT the removed \$149 product.
- `STRIPE_HUNT_PLAN_PRICE_ID` env var is the \$19 parcel unlock Stripe Price ID — KEEP IT.
- `interactive-map.tsx` `onCheckout` prop was removed; 3D Unlock Intel callback now redirects to `/intel?lat=X&lng=Y&address=Z` so users still have a conversion path.
- `lib/state-config.ts` `quick_look` field is a legacy flag — product discontinued but field kept for backward compatibility.

Orphaned Marketing Copy (Out of Scope — user noted for follow-up)

- \$350 Full Report and \$79 Hunting Intel references in `/brokers/page.tsx` and `/intel/page.tsx` — marketing copy, not wired to any active product.

User Preferences

- User (Clark, cwc2030@gmail.com, admin role) prefers surgical/explicit edits, not broad refactors.
- Always ask for confirmation before making changes to auth, billing, DB schemas.
- The user is about to take the first real revenue on this app — stability is paramount.

Package Management

- yarn only (never npm/npx). `yarn prisma generate` for Prisma.

Deployment

- Primary custom domain: `terrafirma.partners`
- All hostnames share a common tag — one `deploy_nextjs_project` call updates all.

Listings (Lease Marketplace) — Chunk 1 shipped Apr 26, 2026

Landowner-side draft form. Full spec & manual test steps in `nextjs_space/docs/listings.md`.

Schema (additive, applied via `prisma db push`)

- model `Listing` — anchored to `SavedProperty` via `savedPropertyId` FK (NOT a Report — no Report model exists in TFP). FK also to `User` via `ownerUserId`.
- Snapshot fields (all nullable): `state`, `county`, `acres`, `terrainScore`, `primaryMovement`, `bedAcres`, `funnelCount`. Pulled from `SavedProperty`.
- Drift tracker: `savedPropertyUpdatedAt` (compare against current `SavedProperty.updatedAt`).
- Lease fields: `askingPriceMin/Max`, `leaseType` (enum), `huntersMax`, `seasonAvailability` (string[]), `amenities` (json).
- Content: `title`, `description` (Text), `photos` (string[]).
- Contact: `contactMethod` (enum), `contactEmail`, `contactPhone`.
- Status: `status` enum (default DRAFT), `publishedAt?`.
- Indexes: `(ownerUserId, status)` + `(state, status)`.
- Enums: `ListingStatus` { DRAFT PENDING_REVIEW PUBLISHED LEASED EXPIRED WITHDRAWN }, `LeaseType`, `ContactMethod`.

OPSEC (HARD CONSTRAINT)

- Listing NEVER stores precise location. NO `centroidLat`, NO `centroidLng`, NO polygon, NO parcel ID column on Listing.
- `state` + `county` only — must match the public county-only map.
- 2 automated tests guard this (`__tests__/listing-opsec.test.ts`):
 1. Schema regex check on `prisma/schema.prisma` Listing block — fails if it sees `centroid|lat|lng`.
 2. Source check on `lib/listings.ts` `snapshotFromSavedProperty()` — same regex.
- If anyone adds lat/lng columns to Listing, the test suite breaks. DO NOT bypass these tests.

API

- `GET /api/listings` — 401 if unauth. List owner's listings.

- `POST /api/listings` — 401 if unauth, 404 if not owner of `SavedProperty`. Body: `{ savedPropertyId }`. Creates DRAFT.
- `GET /api/listings/[id]` — 401/404. Returns listing + `savedProperty` include.
- `PATCH /api/listings/[id]` — 401/404. **409 if status !== DRAFT** (DRAFT-only updates in chunk 1).
- All routes use `Prisma.JsonNull` for amenities `null` translation (zod may produce `null`, Prisma JSON column requires `Prisma.JsonNull`).

Pages

- `/listings` — server, lists owner's listings. 307 → `/login` if unauth.
- `/listings/new` — Step 1 (pick `SavedProperty`). Redirects to `/listings` if owner has 0 `SavedProperties`.
- `/listings/[id]/edit?step=1|2|3` — wizard resume.

Tests

- `yarn test` — 43/43 passing (2 OPSEC + 28 zod validation + 13 API integration). vitest config loads `.env` for DB connection.
- `vitest@4.1.5` + `@vitest/coverage-v8` added as dev deps.

Asset filter gotcha (env-level)

- The file-write layer auto-replaces `.jpg / .png` URL substrings inside `.ts / .tsx` strings with real CDN URLs (often produces unterminated strings).
- Workaround: write file via `python3 /tmp/script.py` heredoc → bypasses filter.
- For test fixtures with URLs, use `'https://cdn.example.com/listings/' + 'photo1' + '-asset'` (no image extension) to avoid filter triggering.

Chunk 3 — Public marketplace + lifecycle (shipped Apr 26, 2026)

Routing

- Owner-side pages MOVED: `/listings/*` → `/dashboard/listings/*`. API paths (`/api/listings/*`) UNCHANGED.
- Public marketplace lives at `/listings/*`:
- `GET /listings` — paginated grid, filters via search params (`state/grade/leaseType/season/acresMin-Max/priceMin-Max`), sort: `newest|highScore|lowPrice|largestAcres`. PUBLIC, no auth.
- `GET /listings/[slug]` — public detail. Slug format `<state>-<county>-<acres>ac-<grade>-<lease>-<cuid>`. Cosmetic; lookup happens by trailing `cuid` via `extractIdFromSlugId()`. 404 unless status = `PUBLISHED`. Canonical-redirects when slug drifts.
- `GET /listings/[slug]/inquire` — placeholder ("Inquiries open soon"). Real flow lands chunk 4.

Lifecycle API (all auth-gated, owner-only)

- `POST /api/listings/[id]/publish` — DRAFT → `PENDING_REVIEW` (or `PUBLISHED` if `TFP_LISTINGS_AUTO_APPROVE=true`, default true). Refreshes snapshot from `SavedProperty`. Validates with `validateForPublish()`. 400 on missing required fields.
- `POST /api/listings/[id]/withdraw` — DRAFT|`PENDING_REVIEW`|`PUBLISHED` → `WITHDRAWN`.
- `POST /api/listings/[id]/lease` — `PUBLISHED` → `LEASED`.
- `POST /api/listings/[id]/relist` — `LEASED` → `PUBLISHED` + new `publishedAt`.
- All return 401/403/404/409 cleanly. Owner-row UI calls these from `<ListingRowActions />`.

lib/listings.ts additions

- `validateForPublish(c)` — field-level publish validation. Returns `PublishValidationError[]`.

- `snapshotFromSavedProperty(sp)` — returns ONLY `{acres, terrainScore, primaryMovement, bedAcres, funnelCount, savedPropertyUpdatedAt}`. Does NOT touch state/county (SavedProperty has no such columns). state/county are owner-supplied via wizard PATCH.
- `listingSlug(l)` — kebab from state-county-acres-grade-leaseType.
- `extractIdFromSlugId(slugId)` — extracts trailing cuid via `/^c[a-z0-9]{24,}$/`.
- `stripForPublic(o)` — recursive OPSEC filter, strips keys matching `/lat|lng|long|geom|polygon|address|coord|parcel/i`. Belt-and-suspenders for any future leak.
- `gradeMinScore(grade)` — inverse of `gradeFromScore` for filter bar.
- `updateListingSchema` adds `state` (regex `/^[A-Z]{2}$/`) and `county` (1-80 chars), settable via wizard PATCH.

lib/county-centroids.ts (NEW)

- Static MO county centroids (114 counties at 4-decimal ~11m precision) + `US_STATE_CENTROIDS` fallback for KS/IA/IL/AR/OK/TN/KY/NE/TX/WI/MN/IN/OH/PA/NY/MI.
- `lookupCentroid(state, county) → Centroid | null` where `Centroid = {lat, lng, precision: 'county' | 'state'}`.
- Drives `<CountyMap />` static-map renderer at zoom 7 (county) / 5 (state) only — never parcel-level.
- Values are public, low-precision, NOT derived from any user data (compliant with OPSEC).

Public Marketplace Components

- `app/listings/_components/listing-card.tsx` — server card.
- `app/listings/_components/filter-bar.tsx` — client filter UI driving URL params.
- `app/listings/[slug]/_components/grade-badge.tsx` — Certificate-style A+ block (gold/dark-emerald serif on cream, thin gold border).
- `app/listings/[slug]/_components/county-map.tsx` — Mapbox static image, county-level only.
- `app/listings/[slug]/_components/photo-gallery.tsx` — lazy-loaded, max 6 photos.

OPSEC defense in depth

- Public detail page `select` s an explicit allowlist (no savedProperty include).
- Detail page passes result through `stripForPublic()` before render.
- No `lat|lng|long|geom|polygon|address|coord|parcel` field name exists anywhere in the response chain.

Env flags

- `TFP_LISTINGS_AUTO_APPROVE` — defaults `true` (MVP). Reads case-insensitive `true|1|yes`. NOT set in `.env`; rely on default. Set explicitly to `false` once admin moderation queue ships in chunk 4.

Navbar

- Added “Marketplace” link to desktop + mobile nav, between “Map Tool” and “Pricing”. Active when path starts with `/listings`.
- Dashboard page got a “My Listings” button next to “New Report”.

Schema migration

- NONE. All chunk-3 columns (`state`, `county`, etc.) already existed from chunk 1.

Tests (must-haves DONE Apr 26, 2026)

- `__tests__/listing-snapshot.test.ts` (5 tests) — pure unit on `snapshotFromSavedProperty`. Asserts no `lat|lng|long|geom|polygon|address|coord|parcel` keys, fields map correctly, null fields pass through, output JSON contains no `centroidLat/centroidLng` numeric values.

- `__tests__/listing-lifecycle.test.ts` (9 tests) — covers ownership (User B → 403), status guards (DRAFT publish missing fields → 400 with field errors; DRAFT lease → 409; DRAFT relist → 409; already-PUBLISHED publish → 409; DRAFT withdraw → 200; unauth → 401) AND auto-approve flag (`TFP_LISTINGS_AUTO_APPROVE=true` → PUBLISHED + `publishedAt`; `=false` → PENDING_REVIEW, no `publishedAt`).
- `__tests__/listing-public-opsec.test.ts` (9 tests) — CRITICAL. Renders the actual `/listings/[slug]` server component to HTML via `react-dom/server.renderToStaticMarkup`. Asserts no forbidden tokens (centroidLat/centroidLng/polygon/geometry/parcels), no source SavedProperty `centroidLat/centroidLng` numeric values (rounded to 4/5/6 decimals), and no match for the lat/lng pair leak regex `/-?\d{1,3}\.\d{4,}\. *-?\d{1,3}\.\d{4,}/`. Plus status-gate: DRAFT/PENDING_REVIEW/WITHDRAWN/LEASED → `notFound()`; PUBLISHED → renders.
- LEASED public behavior decision: DOCUMENTED. `loadPublished()` filters `status: 'PUBLISHED'` so LEASED listings 404 from public surfaces — keeps prospects from inquiring on already-taken inventory.
- `vitest.config.ts` now sets `oxc: { jsx: { runtime: 'automatic' } }` to override Next's `tsconfig.jsx: 'preserve'` so `.tsx` page imports transform in tests.

Test totals

- 96/96 passing (was 73 baseline + 5 snapshot + 9 lifecycle + 9 OPSEC = 96).

What was DELIBERATELY deferred to chunk 3b (NOT done in MVP)

- Filter behavior on `/listings` (state/grade/acres/price/leaseType/season filters)
- Sort behavior tests
- Slug-format tests
- Pagination tests
- og: metadata tests
- County-map render tests
- Photo upload via cloud storage (still relies on existing `photos: string[]` PATCH).
- Admin moderation queue UI (auto-approve env flag covers MVP).
- Real inquiry flow (placeholder page only).

Chunk 4 — Public Inquiry Flow (shipped Apr 26, 2026)

End-to-end hunter→landowner inquiry funnel on PUBLISHED listings. Hunter submits form → landowner gets email → hunter gets confirmation → owner sees in `/dashboard/inquiries`.

Schema

- `model Inquiry` — anchored to Listing via `listingId` FK, NO User FK on the hunter side (hunters submit publicly without accounts).
- Migration: `prisma/migrations/20260426154403_add_inquiry/migration.sql`.
- Fields: `id`, `listingId` (FK), `hunterEmail` (lowercased), `hunterName`, `hunterPhone?`, `preferred-Dates?`, `partySize` (default 1), `message` (Text), `status` (default NEW), `source?`, `utmSource/Medium/Campaign?`, `ipHash?`, `createdAt`, `repliedAt?`, `closedAt?`.
- Enum `InquiryStatus` { NEW REPLIED CLOSED }.
- Indexes: `[listingId, status]`, `[hunterEmail]`.
- Listing back-relation: `inquiries Inquiry[]` added.

OPSEC continuation

- Inquiry stores hunter contact + message ONLY. No location columns.

- Email bodies use `state/county/acres/lease/price` ONLY via `EmailListingContext` interface. The interface enforces this at the type level — emails CANNOT receive `lat/lng/coords`.
- Unit tests guard: `__tests__/inquiry-api.test.ts` “OPSEC continuation” group inspects rendered email HTML, asserts no `centroidLat|centroidLng|latitude|longitude|parcel|address|geom|polygon|coordinate` substrings, no `lat/lng` decimal patterns.
- `inquiryInputSchema` is `.strict()` — rejects unknown fields like `centroidLat`, `lat`, `lng`, `coords`, `parcel`, `address`, `geom`.

Helpers

- `lib/email.ts` — Abacus `sendNotificationEmail` wrapper. Default `abacusTransport`; `setEmailTransport(stub)` / `resetEmailTransport()` for test injection. `sendEmail({to, subject, html, bcc?, replyTo?})` — BCC is a 2nd API call (Abacus accepts one recipient per call). `replyTo` not on the wire (Abacus doesn’t support it); landowner email body shows hunter email as a `mailto:` link.
- `lib/inquiry.ts` — `inquiryInputSchema` (strict zod), `ipHash()`, `resolveClientIp(headers)` (left-most x-forwarded-for → x-real-ip), `landownerSubject()` / `buildLandownerEmail()`, `hunterSubject()` / `buildHunterConfirmationEmail()`. Constants `IDEMPOTENCY_WINDOW_MS=7d`, `RATE_LIMIT_MAX=3`, `RATE_LIMIT_WINDOW_MS=1h`. `esc()` HTML-escapes via `replace(/.../g, ...)` (NOT `replaceAll` — TS target=es2020).

API

- `POST /api/listings/[id]/inquire` — PUBLIC, no auth. Order: parse → validate → honeypot silent drop → listing PUBLISHED-only 404 → rate-limit 429 → idempotency 7d → create → fire emails (errors logged, never 500). Returns `{ok:true, inquiryId, slug, idempotent?}`. GET returns 405.
- `POST /api/dashboard/inquiries/[id]/status` — owner-only. Body `{status: "REPLIED"|"CLOSED"}`. 401/404(collapsed: not-found OR not-yours)/400/409(already-target)/200. Stamps `repliedAt` / `closedAt`.

Pages

- `/listings/[slug]/inquire` — REPLACED chunk-3 placeholder. Server-validates PUBLISHED, renders header + `<InquiryForm />`. 404 for any non-PUBLISHED.
- `app/listings/[slug]/inquire/_components/inquiry-form.tsx` — client form. Posts to `/api/listings/{listingId}/inquire`. Honeypot field `website` (off-screen). Pulls UTM’s from URL. Fires `trackEvent('listing_inquiry', ...)`. 429/404 surface as user-friendly errors.
- `/dashboard/inquiries` — auth-gated, redirects to `/login?callbackUrl=%2Fdashboard%2Finquiries`. Table: Date | Listing | Hunter | Status | Actions. Newest first. Scoped to owned listings.
- `_components/inquiry-row-actions.tsx` — NEW→[Mark Replied, Close], REPLIED→[Close], CLOSED→none. POSTs to status route, calls `router.refresh()` on success.
- `_components/inquiry-message-disclosure.tsx` — 140-char preview + Show more/less.
- Dashboard “Inquiries” link added to `/dashboard` between “My Listings” and “New Report”.

Email contract

- **Landowner inbox fallback chain:** `Listing.contactEmail` → `User.email` → `console.warn` and skip (inquiry still saved, dashboard still shows it; no crash).
- BCC: `INQUIRY_BCC=clark@terrafirma.partners` env var. ALL landowner emails BCC clark for ops visibility.
- Hunter confirmation copy: subject “We forwarded your inquiry – {county}, {state}”, “what happens next” body with 24-72hr reply window, no listing-detail link back.

Status transitions

- NEW → REPLIED (stamps `repliedAt = now`).

- `NEW` → `CLOSED` (stamps `closedAt = now`).
- `REPLIED` → `CLOSED` (stamps `closedAt = now`).
- 409 if already in target.
- Returning to `NEW` is NOT exposed via this endpoint by design.

Idempotency / rate-limit

- Idempotency: same (`listingId`, `hunterEmail`) within 7d returns existing `inquiryId`, `idempotent: true`, NO email re-fire.
- Rate limit: max 3 per hour per `ipHash`. 429 + `Retry-After` header. Salted via `INQUIRY_IP_SALT` (sha256). Raw IPs NEVER stored.

Env vars added

- `INQUIRY_BCC=clark@terrafirma.partners`
- `INQUIRY_IP_SALT=<sha256 hex>` (set once via `openssl rand -hex 32`).

Tests (chunk 4) — 40 new

- `__tests__/inquiry-validation.test.ts` (17 tests) — pure zod. Strict reject (`centroidLat/lat/lng/coords/parcel/address/geom`), required fields, email lowercasing, message 20-2000 bounds, partySize 1-50, honeypot accepted, phone min len.
- `__tests__/inquiry-api.test.ts` (23 tests) — mocks `next-auth`, injects `setEmailTransport(captureTransport())`. Covers status guard (`DRAFT/PENDING_REVIEW/LEASED/WITHDRAWN` → 404 no leak, 0 emails, 0 rows), idempotency 7d (same `inquiryId`, no re-fire), rate-limit 3/hr (429 + `Retry-After`, leftmost xff IP, `ipHash` stored not raw), email firing (3 calls: `landowner-To` + `landowner-BCC` + `hunter`), owner-inbox fallback (`contactEmail` → `owner.email` → `warn+skip`), OPSEC keys/decimals, honeypot silent 200, status route 401/404/200/409/400.

Test totals

- 136/136 passing (96 baseline + 17 validation + 23 API).

Chunk 5 / 4b backlog

- Admin moderation queue (set `TFP_LISTINGS_AUTO_APPROVE=false`, gate publish through admin review).
- Waitlist admin dashboard + CSV export.
- Photo upload via cloud storage.
- Filter/sort/slug/pagination/og/county-map render tests (chunk 3b).
- Inquiry filter/sort UI on `/dashboard/inquiries`.
- Inquiry double-opt-in email confirmation (catch invalid hunter emails before notifying landowner).

Waitlist (FB Landing Pages) — Chunk 2 shipped Apr 26, 2026

Two **public, no-auth** landing pages to convert FB ad traffic into typed leads while the marketplace is being built. Full spec in `nextjs_space/docs/waitlist.md`.

Schema

- `model Waitlist` — single shared table for both sides. PK `id`, `side` (`LANDOWNER|HUNTER`), `email`, optional `name/state/states[]/acres/maxBudgetUsd/seasonInterest[]/groupSize/source/utmSource/utmMedium/utmCampaign/notes`, `createdAt`.
- Indexes: (`side`, `state`) and (`email`).

- Enum: `WaitlistSide { LANDOWNER HUNTER }`.
- **(email, side) is the dedupe key** — second submission updates the existing row in place.

Migration system NOW USES `prisma migrate dev`

- Baseline created via `prisma migrate diff --from-empty --to-schema-datamodel ... --script > baseline.sql` saved to `prisma/migrations/0_init/migration.sql` and resolved as applied.
- This migration: `prisma/migrations/20260426141649_add_waitlist/migration.sql`.
- **Going forward use `yarn prisma migrate dev --name <change>` for any schema change** instead of `db push`. Migration history is now committed.

OPSEC

- `waitlistInputSchema` is `.strict()` — unknown fields like `centroidLat` are rejected with 400 before they reach the DB. Same hardening pattern as Listing.

API

- `POST /api/waitlist` — public, no auth.
- 201 + `{ id, mode: "created" }` on new row, 200 + `{ id, mode: "updated" }` on idempotent update.
- 400 on validation, 500 on DB error.
- UTM resolution: body → request URL query → Referer header. First non-empty wins per key.
- No GET/PATCH/DELETE exposed publicly.

Pages

- `/lease-your-land` — landowner. Hero IS the form. Below: how-it-works → why-list → FAQ. `source = "lease_your_land_landing"`.
- `/find-a-lease` — hunter. Hero IS the form (multi-state, season chips). Below: certified-explainer → how-it-works → FAQ. `source = "find_a_lease_landing"`.
- Both fire GA4 `waitlist_join` on success, forward UTMs from URL, show inline thank-you/error.
- Forms live in `app/<page>/_form/*.tsx`, shared helpers in `app/_landing-shared/*` — `_` prefix excludes from routing in Next.js App Router.

Design

- Hero `bg-gradient-to-br from-stone-800 via-emerald-900 to-stone-900` matches homepage hero (FB ad continuity).
- Below-fold sections alternate `bg-stone-50 / bg-white`.
- Form is white card on dark hero (`bg-white rounded-2xl shadow-2xl`).
- Buttons: `bg-emerald-700 hover:bg-emerald-800` (matches pricing/navbar primary CTA).

Tests

- `yarn test` — 73/73 passing total. Waitlist adds 30 new tests: 19 zod validation + 11 API integration.
- vitest config auto-loads `.env`.

Asset filter workaround (re-confirmed)

- File-write layer still auto-replaces `.jpg / .png` strings in `.ts / .tsx` files. Bypass via `python3 /tmp/script.py heredoc`.
- Python heredoc gotcha: don't escape `$` (Python doesn't need it) — `\$` inside JS template strings becomes literal `\$` and breaks parser. Use `$disconnect` directly in heredoc.

NOT in Chunk 2 (planned)

- Chunk 3: photo upload + public listings discovery + admin/lifecycle.
- Chunk 4: waitlist admin dashboard, CSV export, double-opt-in email confirmation.